

Perception Stack: Grounding Vision-Language Models with Structured Physical Perception

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Abstract—Vision-language models (VLMs) excel at object recognition and semantic understanding but lack grounding in the physical properties necessary for temporal and causal reasoning in video. A VLM presented with a single video frame has no access to object depth, inter-frame motion, or collision dynamics; it must infer these properties implicitly from visual appearance alone. We propose Perception Stack, a four-stage pipeline that extracts structured physical measurements from video using specialist perception models and injects them as natural language context into a VLM before it answers physics-grounded questions. The pipeline comprises object segmentation via SAM 2, monocular depth estimation via Depth Anything V2, optical flow estimation via RAFT, and context-augmented question answering via LLaVA-1.5 7B. Our central hypothesis is that a VLM provided with explicit physical grounding, including object trajectories, collision-plane depth estimates, and velocity vectors, should substantially outperform a VLM operating on image input alone. We evaluate this hypothesis on 70 clips from the CLEVRER benchmark across four question types: descriptive, explanatory, predictive, and counterfactual. The pipeline requires no model fine-tuning, no architectural modifications, and operates entirely from pretrained checkpoints.

Index Terms—vision-language models, physical reasoning, context injection, video understanding, optical flow, depth estimation, object segmentation

I. INTRODUCTION

Physical reasoning in video, the capacity to predict object trajectories, infer causal relationships between collision events, and reason about counterfactual outcomes, represents one of the most challenging frontiers in visual understanding. Human observers accomplish these tasks effortlessly, drawing on an internalized model of Newtonian dynamics built from years of embodied experience. Computational models, by contrast, have historically relied on either end-to-end learned representations that conflate perception and reasoning, or symbolic physics engines that require precise geometric state as input.

The emergence of large vision-language models (VLMs) has opened a new avenue: models that combine visual perception with powerful language-based reasoning. Yet despite their broad capability, VLMs consistently underperform on tasks requiring physical intuition [1]. The root cause is not model capacity but missing grounding. When a VLM is given a single video frame and asked whether two objects will collide, it has no access to the depth separation between those objects, their

velocities, or their trajectories. It generates an answer from static appearance and language priors alone.

Perception Stack addresses this grounding gap through structured context injection. Rather than retraining or fine-tuning the VLM, we extract physically meaningful measurements using specialist computer vision models and format them as natural language context prepended to each question. The core intuition is that a model told “the green sphere is moving rightward at 5.36 px/frame, depth 0.27, and its mask overlapped with the yellow cylinder at frame 92” is better equipped to reason about subsequent physical events than one given only a static frame.

Our pipeline consists of four stages. Stage 1 detects objects and generates tracking prompts using the CLEVRER Mask R-CNN proposal files. Stage 2 tracks all objects across 128 frames using SAM 2, producing centroid trajectories, entry and exit frames, and mask overlap events. Stage 3 estimates per-object depth at physically significant frames using Depth Anything V2. Stage 4 estimates per-object velocity vectors using RAFT optical flow. The outputs of all stages are assembled into a structured natural language context string and injected into LLaVA-1.5 7B alongside the question.

The primary contributions of this work are as follows. First, we present a modular four-stage physical perception pipeline that produces structured, human-readable context from raw video clips without any model training. Second, we provide a controlled empirical evaluation on the CLEVRER benchmark under five conditions: a baseline with object identity only, a full pipeline condition, and three ablation conditions each removing one pipeline stage. Third, we offer an analysis of the behavioral response of a lightweight VLM to structured physical context injection, including failure modes that motivate future directions for this line of research.

II. RELATED WORK

A. Video Reasoning Benchmarks

CLEVRER [1] is a diagnostic video question answering dataset designed specifically to evaluate temporal and causal reasoning. The dataset comprises 20,000 synthetic video clips of 3D objects interacting on a flat tabletop, generated using the Bullet physics engine and rendered with Blender. Each clip is paired with four question types: descriptive, explanatory,

predictive, and counterfactual. Yi et al. demonstrate that state-of-the-art video QA models perform near random chance on causal question types, establishing that strong visual perception alone is insufficient for physical reasoning. CLEVRER provides ground truth annotation files with per-frame object trajectories and collision events, which we use for pipeline validation and threshold derivation in this work.

B. Vision-Language Models

LLaVA [2] introduces visual instruction tuning, a paradigm in which language-only GPT-4 is used to generate multimodal instruction-following data for training an end-to-end vision-language model. LLaVA connects a CLIP visual encoder to a Vicuna language model via a lightweight projection layer, enabling general-purpose visual question answering from a single image. We use LLaVA-1.5 7B, an improved variant that substitutes a larger MLP connector and trains on a broader instruction mixture. Sa2VA [3] is the most closely related prior work to our approach: it unifies SAM 2 and a multimodal LLM into a shared token space for dense grounded understanding of images and videos. Sa2VA demonstrates that combining segmentation and language reasoning in a jointly trained model yields strong performance on referring video object segmentation. Our work differs fundamentally in that we do not jointly train any component; instead, we use SAM 2 as a perception stage and pass its outputs as language context to a frozen VLM.

C. Specialist Perception Models

SAM 2 [4] extends the Segment Anything paradigm to video, enabling prompted tracking of arbitrary objects across frames via a memory-conditioned architecture. We use SAM 2 base+ to propagate object masks across all 128 frames of each clip from a single point prompt per object.

Depth Anything V2 [5] is a monocular depth estimation foundation model trained on a combination of synthetic labeled data and large-scale pseudo-labeled real images. It produces relative depth maps from single RGB frames. We use the Small variant, which is sufficient for the coarse depth scalars required by our pipeline.

RAFT [6] introduces Recurrent All-Pairs Field Transforms for optical flow estimation. By constructing a 4D correlation volume over all pixel pairs and iteratively refining flow predictions through a recurrent update operator, RAFT achieves state-of-the-art accuracy on standard optical flow benchmarks. We use the small variant accessed via the torchvision model zoo.

III. METHODS

A. Problem Formulation

Let $V = \{f_1, f_2, \dots, f_{128}\}$ denote a video clip of 128 RGB frames at 25 fps. Let Q denote a natural language question and $C = \{c_1, c_2, \dots, c_k\}$ the set of candidate answer choices, where $k \in \{2, 3, 4\}$ depending on question type. The task is to predict the correct subset $C^* \subseteq C$.

We evaluate two conditions. In the **baseline** condition, the model receives the last frame f_{127} , an object identity list extracted from annotations, and the question Q . In the **enhanced** condition, the model additionally receives a structured context string $\mathcal{S}$ derived from the four pipeline stages, described in Sections III-D through III-G.

We report three evaluation metrics. **Per-question accuracy** requires the predicted set to exactly match the ground truth:

$$\text{Acc}_Q = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{C}_i = C_i^*] \quad (1)$$

Per-option accuracy treats each individual choice as an independent binary classification:

$$\text{Acc}_{\text{opt}} = \frac{\sum_{i,j} \mathbf{1}[\hat{y}_{ij} = y_{ij}]}{\sum_i k_i} \quad (2)$$

where $y_{ij} \in \{0, 1\}$ indicates whether choice j of question i is correct.

F1 score measures partial credit for multi-select questions via precision and recall over the predicted and ground truth choice sets:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}, \quad P = \frac{|\hat{C} \cap C^*|}{|\hat{C}|}, \quad R = \frac{|\hat{C} \cap C^*|}{|C^*|} \quad (3)$$

For questions with no correct answers ($|C^*| = 0$), $F_1 = 1.0$ if and only if the model predicts no correct choices, and $F_1 = 0.0$ otherwise.

B. Dataset and Evaluation Protocol

We evaluate on the CLEVRER validation split [1]. From the available 5,000 validation clips (indices 10001 to 15000), we select the first 70 clips in index order. Table I summarizes the question distribution across the 70 clips.

TABLE I: Question distribution across 70 evaluation clips.

Question Type	Questions	Answer Format
Descriptive	770	Single token
Explanatory	129	Multi-select MC
Predictive	50	Single-choice MC
Counterfactual	129	Multi-select MC
Total	1,078	

Descriptive questions are scored by exact string match after vocabulary normalization. Predictive questions are scored by exact letter match. Explanatory and counterfactual questions are scored by all three metrics. An edge case applies when the ground truth has zero correct choices: the model scores $F_1 = 1.0$ only if it outputs “None”; any predicted choice yields $F_1 = 0.0$.

C. Baseline: LLaVA with Object Identity

The baseline condition provides LLaVA-1.5 7B with the last frame of the clip (frame 127, JPEG), an object identity list derived from the CLEVRER annotation files, and the question with answer choices. The model is loaded with 4-bit NF4 quantization via

bitsandbytes, using `bnb_4bit_quant_type='nf4'`, `bnb_4bit_compute_dtype=torch.float16`, and `device_map='auto'` on an NVIDIA A100 GPU.

The prompt follows the LLaVA instruction format with a domain-level pre-context, the object identity block, the question body, and an `ANSWER:` token at the end to prime the model’s generation. Answer parsing uses a three-tier strategy: Tier 1 extracts the answer directly from the `ANSWER:` field; Tier 2 scans the first line of the full output for valid tokens; Tier 4 falls back to an empty prediction scored as incorrect. A vocabulary normalization map corrects common paraphrases (e.g., “round” to “sphere”, “metallic” to “metal”). Letter extraction for multi-select questions uses the regex pattern `\b([a-d])\b` applied to the first output line only, preventing false positives from reasoning sentences.

Object identity is included in both baseline and enhanced conditions. This design decision ensures that the only isolated variable is the physical context derived from the pipeline stages, enabling a controlled comparison.

D. Stage 1: Object Detection and Prompt Generation

Generating accurate point prompts for SAM 2 requires precise object centroids in the first visible frame. We evaluated three approaches before arriving at our final strategy.

Annotation world coordinates were rejected because the CLEVRER annotation files store 3D Bullet physics coordinates without a documented projection matrix mapping them to pixel space.

YOLOv8 (large variant, confidence threshold 0.1) was rejected after empirical evaluation revealed a severe domain gap: COCO-trained YOLO detects zero cube instances across all test clips, misidentifies cylinders as cups or bottles, and produces unreliable sphere detections. The synthetic shiny geometry of CLEVRER objects falls entirely outside the COCO visual distribution.

HSV-based segmentation was rejected due to failure modes under shadow regions, object occlusion, and similar object hues. These conditions are common in CLEVRER clips with multiple same-colored objects.

The final approach uses the CLEVRER **derender proposal files**, produced by a Mask R-CNN trained specifically on CLEVRER frames [1]. Each proposal file contains per-frame RLE-encoded instance masks with associated color, material, and shape labels at confidence scores exceeding 0.999. Centroids are extracted from the decoded masks as:

$$(c_x, c_y) = \left(\frac{\sum_{(x,y) \in M} x}{|M|}, \frac{\sum_{(x,y) \in M} y}{|M|} \right) \quad (4)$$

where M denotes the set of foreground pixels in the decoded mask. The prompt frame for each object is set to $t_{\text{prompt}} = t_{\text{first_visible}} + 15$, capped at frame 127, to ensure the object is fully within the camera view before tracking begins.

E. Stage 2: Object Tracking via SAM 2

SAM 2 base+ [4] is initialized with one point prompt per object at the prompt frame determined in Stage 1. The

predictor propagates masks both forward and backward from the prompt frame to cover all 128 frames.

From the raw mask tensors (shape $128 \times 320 \times 480$), we compute the following derived quantities per object:

Centroid trajectory: Per-frame centroids using Equation (4). Frames where the mask is empty are recorded as NaN.

Object area: $A(t) = \sum_{x,y} M(x,y,t)$, used to infer entry and exit events.

Entry and exit frames: Frame indices where $A(t-1) = 0$ and $A(t) > 0$ (entry), or $A(t-1) > 0$ and $A(t) = 0$ (exit).

Mask overlap: For each ordered pair of objects (i, j) , the first frame where their masks intersect:

$$t_{\text{overlap}}(i, j) = \min \left\{ t : \sum_{x,y} M_i(x, y, t) \cdot M_j(x, y, t) > 0 \right\} \quad (5)$$

Raw mask tensors are not saved to disk due to storage constraints (approximately 60 MB per clip). Only the derived quantities are written to compressed `.npz` files on Google Drive. This design decision introduces an approximation in Stage 4, discussed in Section III-G.

Figure 1 shows representative SAM 2 mask overlays for one evaluation clip.

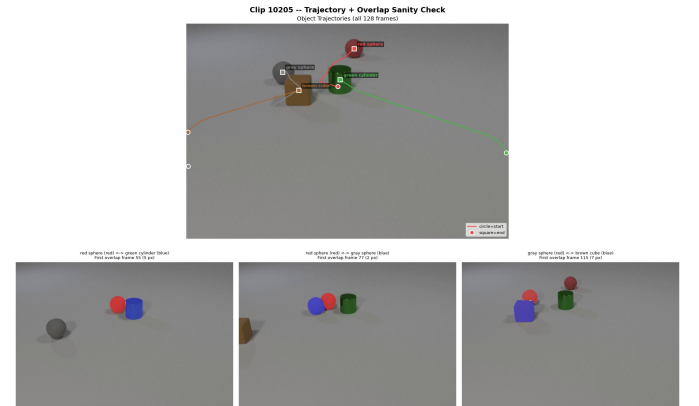


Fig. 1: SAM 2 output for clip 10205. Top: centroid trajectories overlaid on the last frame across all 128 frames. Bottom: RGB frames at the first detected mask overlap for each object pair, with overlap frame index and pixel count annotated.

F. Stage 3: Depth Estimation via Depth Anything V2

Depth Anything V2 Small [5] is applied to selected frames via the HuggingFace pipeline API. We sample only two categories of frames per clip: the overlap frame for each SAM 2 overlap pair, and the last frame (frame 127). This targeted sampling strategy avoids redundant inference on frames with no physically significant events.

For each sampled frame, the raw depth map is normalized per-frame to the range $[0, 1]$:

$$\hat{d}(x, y) = \frac{d(x, y) - d_{\min}}{d_{\max} - d_{\min}} \quad (6)$$

where $d_{\min}$ and $d_{\max}$ are the minimum and maximum depth values in that frame. The normalized depth values are relative estimates valid only within a single frame; cross-frame depth comparisons are not meaningful.

Per-object depth is sampled at the centroid pixel from the SAM 2 output:

$$\hat{d}_{\text{obj}} = \hat{d}(c_x, c_y) \quad (7)$$

A collision-plane threshold τ is derived empirically from the ground truth annotation files. For each confirmed annotation collision event, we retrieve the normalized depth of both participating objects at frame $t_{\text{collision}} - 5$ and compute the absolute difference. Averaging across all confirmed collisions in the evaluation set yields $\tau = 0.08$. Two objects at a spatial overlap frame are classified as co-planar if:

$$|\hat{d}_A - \hat{d}_B| < \tau \quad (8)$$

Figure 2 shows representative depth maps and per-object depth values for one evaluation clip.

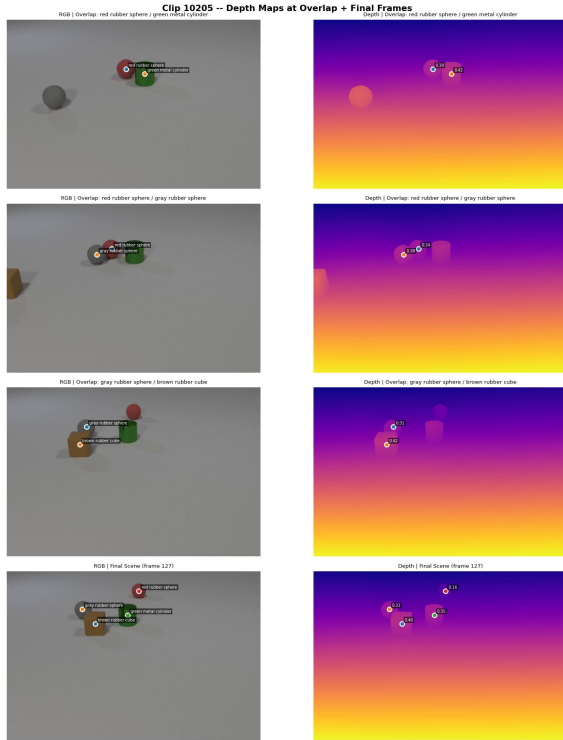


Fig. 2: Depth Anything V2 output for clip 10205. Each row shows the RGB frame (left) and normalized depth map in plasma colormap (right) at one overlap frame and at the final frame (frame 127). Per-object depth scalars are annotated at centroid positions. Depth values are relative within each frame; 0 = closest to camera, 1 = farthest.

G. Stage 4: Optical Flow and Velocity via RAFT

RAFT small [6] is loaded via `torchvision.models.optical_flow.raft_small` with default pretrained weights. Input frames are converted

to float32 tensors in the range $[0, 255]$, matching RAFT’s expected input distribution. CLEVRER frames (480×320) are both divisible by 8, requiring no padding.

Two categories of frame pairs are processed per clip. **Pre-overlap pairs:** for each SAM 2 overlap event involving objects i and j at frame t_{overlap} , we compute flow over the pair $(t_{\text{overlap}} - 3, t_{\text{overlap}})$. A 3-frame gap is used rather than a 1-frame gap to yield displacement magnitudes large enough to classify direction reliably. **Final velocity:** flow over the pair (124, 127) is computed for all tracked objects.

Per-object velocity is estimated by averaging the flow field within a 15×15 pixel window centered at the SAM 2 centroid:

$$v_x = \frac{1}{|W|} \sum_{(x,y) \in W} \phi_x(x,y), \quad v_y = \frac{1}{|W|} \sum_{(x,y) \in W} \phi_y(x,y) \quad (9)$$

where ϕ_x, ϕ_y are the horizontal and vertical flow components from RAFT’s final prediction iteration, and W is the 15×15 window.

This approach is an approximation necessitated by the storage constraint in Stage 2: because raw SAM 2 masks were not saved, we cannot average flow over the full object mask. The centroid window approach introduces two documented failure modes. First, for small or partially visible objects, the window may sample predominantly background pixels. Second, at overlap frames, centroid drift into the mixed-region boundary produces unreliable velocity estimates; these readings should be interpreted as directional indicators rather than precise measurements.

RAFT output is taken from the final element of the prediction list (the most refined estimate). Direction classification uses a threshold of 0.3 px/frame: objects with $|v_x| < 0.3$ and $|v_y| < 0.3$ are described as stationary.

Figure 3 shows representative optical flow arrows for one evaluation clip.

H. Context String Assembly

The outputs of Stages 2, 3, and 4 are assembled into a structured natural language context string S prepended to the question in the enhanced condition. Natural language formatting is used rather than raw coordinate arrays because LLaVA was trained on natural text corpora; structured numerical arrays fall outside its training distribution.

The context string has four sections: (1) the object identity list (color, material, shape); (2) SAM 2 context including entry frames, visibility counts, and overlap events with frame indices and pixel counts; (3) depth context including per-overlap-pair same-plane classification and final scene depths; and (4) RAFT context including pre-overlap velocities and final velocities with direction descriptions. A full annotated example is provided in Appendix A.

Figure 4 illustrates the complete pipeline architecture, contrasting the baseline condition with the full enhanced condition and showing the cumulative context string assembly across all three perception stages.

Four ablation conditions are evaluated in addition to the full pipeline: **no_sam2** (Sections 2 and 3 of S omitted), **no_depth**

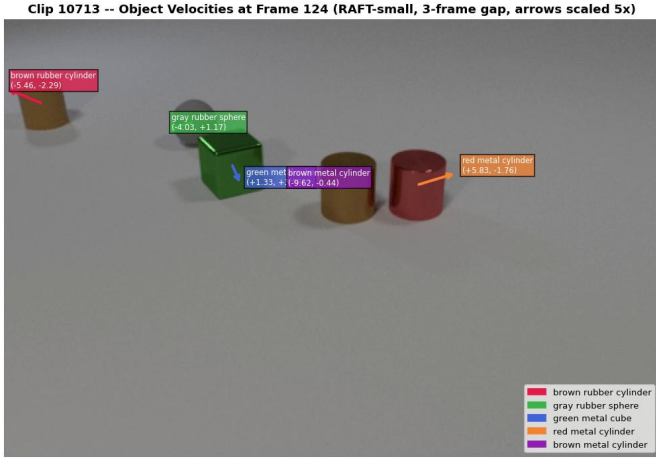


Fig. 3: RAFT-small optical flow for clip 10713 at frame 124. Arrows are scaled $5\times$ for visibility. Per-object velocity vectors (dx , dy) in pixels per frame are annotated at each object centroid. Negative x indicates leftward motion; negative y indicates upward motion.

(Section 3 omitted), and **no_raft** (Section 4 omitted). The object identity block is retained in all conditions.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

All LLaVA inference is run on an NVIDIA A100 40 GB GPU via Google Colab Pro. SAM 2 inference is also run on A100. Depth Anything V2 inference is run on a T4 GPU. RAFT inference is run on A100. All pipeline stages use pretrained checkpoints with no fine-tuning. Table II reports wall-clock inference time per clip for each LLaVA condition.

TABLE II: LLaVA inference runtime per clip across all conditions (70 clips, NVIDIA A100).

Condition	Avg (s)	Min (s)	Max (s)
Baseline	28.7	25.0	37.5
Full	36.1	27.1	47.2
No SAM2	36.8	27.0	48.6
No Depth	32.6	23.2	47.9
No RAFT	29.9	21.9	42.1

The total evaluation comprises $5 \text{ conditions} \times 70 \text{ clips}$, with approximately 15 questions per clip, yielding roughly 5,000 model inference calls across all conditions.

B. Baseline Results

Table III presents the full quantitative results across all conditions and question types. The baseline achieves 0.284 per-question accuracy on descriptive questions, reflecting the inherent difficulty of temporal attribute queries from a single static frame. On explanatory questions, the baseline per-question accuracy of 0.124 is above the near-zero level that would indicate random guessing on a multi-select task. Predictive accuracy of 0.560 is marginally above the 0.500 random

chance level for a binary forced-choice task. Counterfactual per-question accuracy of 0.093 indicates near-random performance on hypothetical reasoning.

C. Full Pipeline versus Baseline

Figures 5 and 6 compare the baseline and full pipeline conditions across all question types.

On descriptive questions, the full pipeline achieves a marginal gain of 0.007 per-question accuracy over the baseline. On counterfactual questions, the pipeline achieves a gain of 0.016. On explanatory questions, per-question accuracy decreases by 0.054, while per-option accuracy increases by 0.015, suggesting the pipeline shifts individual option classifications without improving whole-question prediction. On predictive questions, the baseline outperforms the full pipeline by 0.080. The delta per-question chart is shown in Figure 7.

D. Ablation Study

Figures 8 through 10 present the ablation results across all five conditions.

Across all ablation conditions, performance on each question type remains tightly clustered. No single stage removal produces a consistent, large-magnitude change in any metric. On predictive questions, the no_depth condition (0.540) most closely matches the baseline (0.560), while all other conditions fall below. On counterfactual questions, the full condition achieves the highest per-question accuracy (0.109), marginally outperforming the no_depth condition (0.093) and the baseline (0.093). The F1 ablation reveals that removing any stage degrades F1 on both explanatory and counterfactual question types relative to the already-below-baseline full condition.

E. Question Outcome Breakdown

Figure 11 presents the per-question outcome breakdown categorized into four groups: *pipeline fixed* (baseline wrong, enhanced correct), *both correct*, *both wrong*, and *pipeline broke it* (baseline correct, enhanced wrong).

On descriptive questions, the pipeline fixes 89 questions and breaks 84, for a net gain of 5. The “both wrong” category (462) dominates, indicating that the majority of descriptive questions are answered incorrectly by both conditions. On predictive questions, counts are small due to the 50-question sample, and the differences are not statistically meaningful.

V. DISCUSSION AND SUMMARY

A. Interpretation of Results

The quantitative results do not confirm the central hypothesis that structured physical context injection substantially improves VLM performance on physical reasoning tasks. The observed differences between baseline and full pipeline conditions are small in magnitude and inconsistent in direction across question types. We offer several principled explanations for this outcome.

Model capacity bottleneck. LLaVA-1.5 7B is a lightweight VLM operating under 4-bit quantization. At this scale, the model’s instruction-following capacity is constrained. The



Fig. 4: Perception Stack pipeline architecture. Left: the baseline condition provides LLaVA with the last frame and object identity only. Center: the three perception stages (SAM 2, Depth Anything V2, RAFT) each contribute a structured context block. Right: the full enhanced condition assembles all context blocks into a single natural language string prepended to each question before VLM inference.

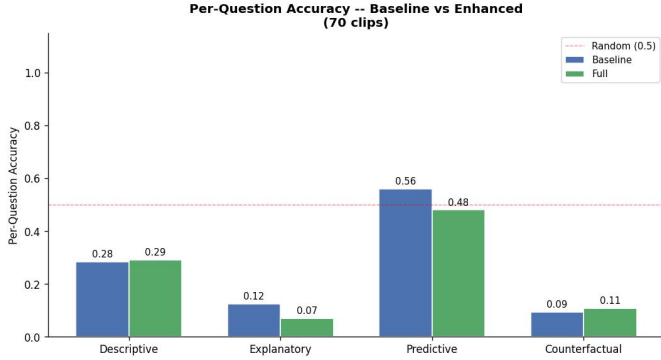


Fig. 5: Per-question accuracy: baseline versus full pipeline across all four question types (70 clips). Dashed line indicates random chance (0.50).

context string we inject is structurally rich, containing object trajectories, depth scalars, same-plane classifications, and velocity vectors. A more capable model, such as LLaVA-1.5 13B, GPT-4V, or a dedicated physics-reasoning VLM, may have the parametric capacity to integrate these structured measurements into coherent reasoning chains. The 7B model appears to extract surface-level keywords from the context rather than performing principled physical inference.

Predictive accuracy at the random baseline. Predictive questions in CLEVRER present exactly two options with exactly one correct answer, making the random chance accuracy 0.50. The baseline achieves 0.560 and the full pipeline

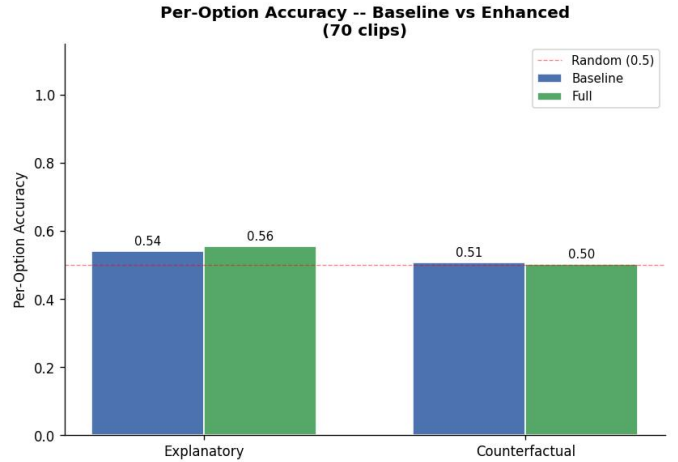


Fig. 6: Per-option accuracy: baseline versus full pipeline for multi-select question types.

achieves 0.480, both within a small margin of chance over 50 questions. Neither condition is performing reliable predictive reasoning; the observed difference reflects stochastic variation around the random baseline rather than a genuine behavioral signal.

Training distribution mismatch. LLaVA-1.5 was trained on natural conversational image-text pairs. Our context string injects structured numerical measurements, for example “depth 0.27”, “velocity +5.36 px/frame”, “mask overlap at frame

TABLE III: Full quantitative results across all conditions and question types (70 clips). Per-Q = per-question accuracy. Per-Opt = per-option accuracy. F1 = macro-averaged F1 score. “–” indicates metric is not applicable for the question type.

Question Type	Condition	Total Q	Per-Q	Per-Opt	F1
Descriptive	Baseline	770	0.284	–	–
	Full	770	0.291	–	–
	No SAM2	770	0.290	–	–
	No Depth	770	0.286	–	–
	No RAFT	770	0.296	–	–
Explanatory	Baseline	129	0.124	0.540	0.497
	Full	129	0.070	0.555	0.427
	No SAM2	129	0.062	0.545	0.416
	No Depth	129	0.054	0.532	0.391
	No RAFT	129	0.062	0.534	0.426
Predictive	Baseline	50	0.560	–	–
	Full	50	0.480	–	–
	No SAM2	50	0.460	–	–
	No Depth	50	0.540	–	–
	No RAFT	50	0.520	–	–
Counterfactual	Baseline	129	0.093	0.507	0.532
	Full	129	0.109	0.502	0.464
	No SAM2	129	0.070	0.482	0.445
	No Depth	129	0.093	0.491	0.396
	No RAFT	129	0.085	0.487	0.484

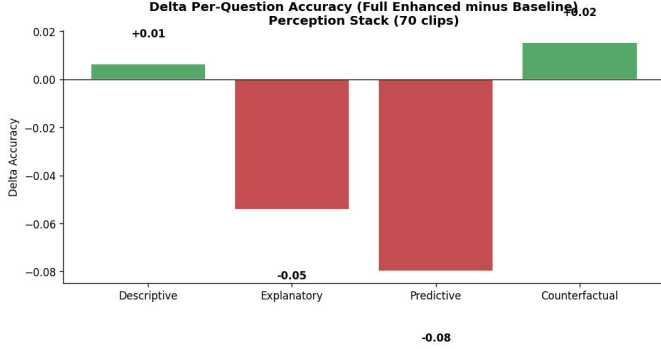


Fig. 7: Delta per-question accuracy (full enhanced minus baseline). Positive values (green) indicate improvement; negative values (red) indicate regression.

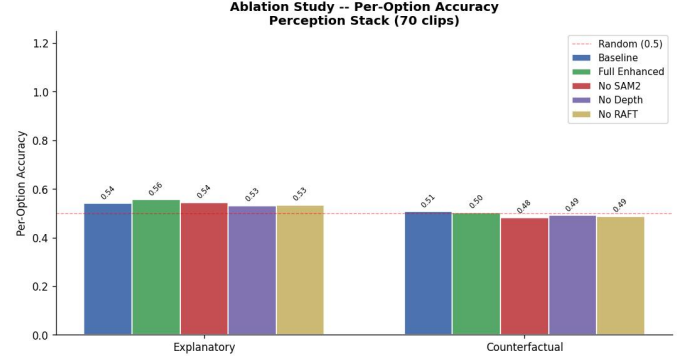


Fig. 9: Ablation study: per-option accuracy for explanatory and counterfactual question types.

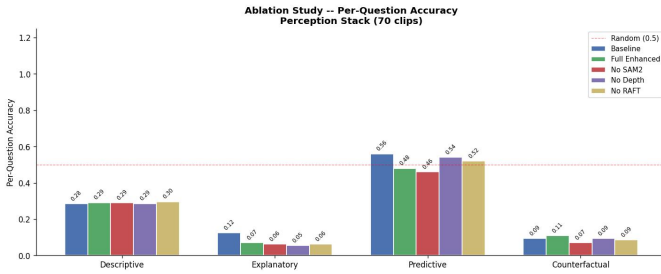


Fig. 8: Ablation study: per-question accuracy across all five conditions and four question types.

92”, that fall outside this training distribution. The model has no learned behavior for integrating per-pixel physical scalars with visual question answering. A model fine-tuned on structured physical context, even briefly, would likely show a substantially stronger response to the same pipeline outputs.

Temporal gap between context and frame. The keyframe presented to LLaVA is always frame 127, the end of the clip, by which point all collisions have already occurred. The context string describes events that happened at earlier frames, which the model cannot observe. The model is asked to reconcile a current static state with historical physical measurements it has no visual access to, which may introduce confusion rather than grounding on question types that require future-state prediction.

Pipeline noise propagation. Each stage introduces approximation error that compounds downstream. SAM 2 centroid estimation depends on proposal file accuracy; depth sampling depends on centroid accuracy; flow sampling depends on both centroid accuracy and mask boundaries. The 15x15 centroid window approximation for flow sampling introduces additional noise for small or partially visible objects. The net effect is that the context string may contain measurements with non-trivial error, particularly at overlap frames where occlusion degrades

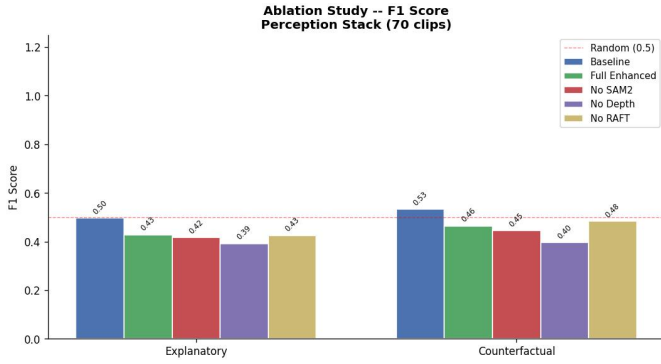


Fig. 10: Ablation study: F1 score for explanatory and counterfactual question types.

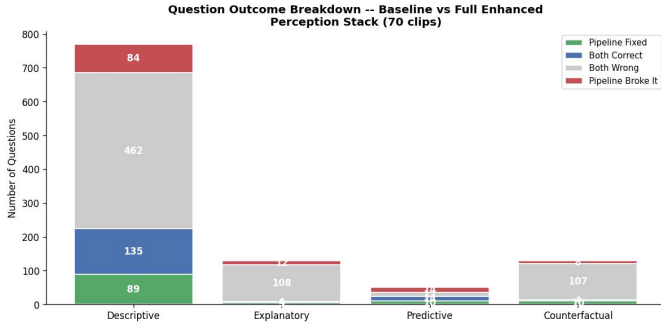


Fig. 11: Question outcome breakdown: baseline versus full pipeline across question types. The dominant category across all types is “Both Wrong”, indicating that neither condition reliably solves most questions.

centroid localization.

Evaluation scale. With 50 predictive questions, 129 explanatory, and 129 counterfactual questions across 70 clips, the evaluation set is not large enough to draw statistically robust conclusions about small accuracy differences. The findings reported here should be interpreted as directional observations rather than definitive measurements.

B. Key Qualitative Finding

The question outcome breakdown in Figure 11 reveals a consistent pattern: the dominant category across all question types is “both wrong”. This indicates that the CLEVRER physical reasoning task, particularly explanatory, predictive, and counterfactual question types, lies substantially above the capability ceiling of LLaVA-1.5 7B regardless of context augmentation. The pipeline shifts the model’s answer distribution in ways that occasionally correct individual questions, but the underlying reasoning mechanism remains keyword extraction rather than principled physical inference.

C. Limitations

Three primary limitations constrain the current evaluation. First, raw SAM 2 mask tensors were not saved due to storage constraints, necessitating the 15x15 centroid window

approximation for flow sampling. Saving full masks would enable more accurate object-region flow averaging. Second, the evaluation is limited to 70 clips; the full CLEVRER validation set of 5,000 clips would yield more statistically reliable conclusions. Third, only one VLM is evaluated; testing larger models (13B, 34B parameter scale) and proprietary VLMs such as GPT-4V would clarify whether the capacity bottleneck hypothesis holds.

D. Future Work

Several directions follow naturally from these findings. Evaluating the same pipeline with a larger VLM backbone would directly test the model capacity hypothesis. Fine-tuning LLaVA on a small dataset of structured physical context pairs would close the training distribution gap identified above. Saving raw SAM 2 masks and performing full-mask flow averaging would improve velocity estimate quality. Scaling evaluation to the full 5,000-clip validation set would enable meaningful statistical analysis. Finally, replacing the centroid-window flow approach with a learned object-region aggregation module may yield more reliable velocity signals.

E. Summary

We have presented Perception Stack, a four-stage pipeline that extracts structured physical measurements from video using specialist perception models and injects them as natural language context into a vision-language model. The pipeline produces object trajectories, relative depth estimates, same-plane collision classifications, and per-object velocity vectors from raw video clips without any model training. Evaluation on 70 CLEVRER clips demonstrates that the pipeline successfully augments the VLM’s input context with physically meaningful information. The central finding is that the capacity of a lightweight VLM to leverage such information at inference time, without fine-tuning and without exposure to structured physical context during training, is the primary bottleneck. These results motivate the exploration of larger VLM backbones and targeted fine-tuning as directions for realizing the full potential of structured physical context injection.

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APPENDIX

A. Baseline Prompt

The following prompt illustrates the baseline condition for an explanatory question. The object identity block is derived from the CLEVRER annotation file. No physical context is provided.

USER: <image>

You are analyzing a single frame from a short video clip. The video depicts a physical scene with objects that may interact over time. Answer the following question based on what you can observe.

Objects in this scene:

- Brown metal cube
- Gray metal cube
- Green metal sphere
- Yellow metal cylinder
- Yellow rubber sphere

Question: Which of the following is not responsible for the metal sphere's colliding with the cylinder?

- a) the collision between the metal sphere and the gray object
- b) the presence of the rubber sphere
- c) the yellow sphere's entering the scene
- d) the presence of the gray object

There may be more than one correct answer, or none.

ANSWER: [one or more letters comma-separated e.g. a, b -- or 'None' if none apply]

REASONING: [2 sentences max]

ANSWER:

Baseline model output:

b, c

REASONING: The rubber sphere and the yellow sphere entering the scene are not responsible for the metal sphere colliding with the cylinder based on visual observation alone.

B. Enhanced Prompt

The following shows the same question under the full pipeline condition. The context string prepends structured physical measurements before the question body.

USER: <image>

You are analyzing a single frame from a short video clip. The video depicts a physical scene with objects that may interact over time. You have been provided with structured physical measurements extracted from the full video. Use this information to answer the following question.

Objects in this scene:

- Brown metal cube
- Gray metal cube
- Green metal sphere
- Yellow metal cylinder
- Yellow rubber sphere

Object overlaps detected: 2

gray cube <-> green sphere first overlap at frame 99 (1 pixels)
green sphere <-> yellow cylinder first overlap at frame 92 (1 pixels)

Depth information (relative estimates, 0 = closest, 1 = farthest).

Threshold: diff below 0.08 = same plane.

Depth at overlap events:

- Gray cube and green sphere at frame 99: depths 0.25 and 0.26, diff 0.01 (same collision plane).
- Green sphere and yellow cylinder at frame 92: depths 0.29 and 0.34, diff 0.05 (same collision plane).

Final scene depth (frame 127):

- Brown metal cube: depth 0.58
- Gray metal cube: depth 0.25
- Green metal sphere: depth 0.27
- Yellow metal cylinder: depth 0.33
- Yellow rubber sphere: depth 0.30

Velocity information (px/frame, 3-frame gap):

Positive x = rightward, negative x = leftward.

Positive y = downward, negative y = upward.

Velocity before overlap events:

- Frame 99: Gray cube (+5.53, -1.77), green sphere (+4.94, -1.90).
- Frame 92: Green sphere (+7.83, -1.05), yellow cylinder (+0.99, +3.19).

Final velocities (frames 124 to 127):

- Brown metal cube: (-9.74, +1.98)
- Gray metal cube: (+5.23, -0.76)
- Green metal sphere: (+5.36, +0.21)
- Yellow rubber sphere: (+2.96, +3.53)
- Yellow metal cylinder: (+1.87, -0.70)

Question: Which of the following is not responsible for the metal sphere's colliding with the cylinder?

- a) the collision between the metal sphere and the gray object
- b) the presence of the rubber sphere
- c) the yellow sphere's entering the scene
- d) the presence of the gray object

There may be more than one correct answer, or none.

ANSWER: [one or more letters comma-separated e.g. a, b -- or 'None' if none apply]

REASONING: [2 sentences max]

ANSWER:

Enhanced model output:

d

REASONING: The presence of the gray object is responsible for the metal sphere's colliding with the cylinder.

Ground truth: a, d (both choices are correct answers).

The baseline predicts b, c ($F1 = 0.00$); the enhanced pipeline predicts d ($F1 = 0.667$), demonstrating partial improvement in identifying one correct answer while missing the other.